Friday Session Report

2026-03-13 01:32:52

Total questions: 5

Question: why blackholes are created in space and why they exists ?

Answer: Black holes are not mysterious “things that just appear”; they are a direct consequence of Einstein’s equations when a large amount of mass is squeezed into a region that is smaller than its Schwarzschild radius. * **How they are created** – When a massive star runs out of nuclear fuel its internal pressure can no longer support it against its own gravity. The star’s surface then follows a trajectory just like a particle that is dropped into a hole: it collapses inward until it passes the radius $r_{S}=2GM/c^{2}$. At that moment a **trapped surface** forms and all of the star’s material is forced toward the central singularity. The exterior spacetime outside the collapsing surface is described by the Schwarzschild metric, and once the surface has entered region II (the black■hole region) a horizon appears from which light cannot escape. * **Why they exist** – The Schwarzschild solution (and more general solutions of Einstein’s equations) predict that if a mass M is concentrated inside a region whose circumference in any direction is smaller than $2\pi r_{S}=2\pi(2GM/c^{2})$, a horizon must form. This is the “Kip Thorne conjecture” quoted in the notes: whenever the mass is packed tightly enough, a black■hole region comes into existence. The equations themselves allow such a solution, and the physical process of stellar collapse provides the mechanism that actually creates it in the universe. Thus, black holes are created when ordinary matter collapses past its Schwarzschild radius, and they exist because the mathematics of general relativity (the Einstein field equations) admit a horizon■forming solution under those conditions. The regions III and IV of the full Schwarzschild diagram are merely mathematical extensions; in realistic collapse they never form, so the observable black hole consists of the exterior region and the interior singularity bounded by the horizon.

Sources:

- Page 76, Asimov Black Holes - The Collapsing Universe.pdf
- Page 77, Asimov Black Holes - The Collapsing Universe.pdf
- Page 67, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 85%

Question: how first propose about the blackhole ?

Answer: I don't have enough information in the notes.

Sources:

- Page 92, Asimov Black Holes - The Collapsing Universe.pdf
- Page 38, Asimov Black Holes - The Collapsing Universe.pdf
- Page 81, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 84%

Question: who first propose about the blackhole ?

Answer: The idea of black holes dates back to the 1700 s, when Pierre Simon Laplace, John Mitchell, and other early thinkers first proposed the concept of a massive “dark” star whose gravity could trap light.

Sources:

- Page 92, Asimov Black Holes - The Collapsing Universe.pdf
- Page 23, Asimov Black Holes - The Collapsing Universe.pdf
- Page 81, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 83%

Question: who is the modern day scientist who's working on blackholes

Answer: I don't have enough information in the notes.

Sources:

- Page 84, Asimov Black Holes - The Collapsing Universe.pdf
- Page 80, Asimov Black Holes - The Collapsing Universe.pdf
- Page 23, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 80%

Question: how does a blackhole looks like

Answer: According to the notes, an observer who hovers near a black■hole would see a **dark “black■plane’’** filling part of the sky – the region from which no light can escape. All of the rest of the universe is bent around this dark patch by the strong curvature of space■time. As a result: * Light from stars that lie behind the hole is lensed around it, so the **entire celestial sphere is projected onto the visible half of the sky**. * The observer sees **multiple concentric images of himself** (or of any object) – a first image at a distance of about $3\pi r_{S}$, a second image at twice that distance, a third at three times, and so on, extending forever. * When the observer is at a radius larger than $1.5\,r_{S}$ the black■hole occupies **less than half of the sky**, while the lensed stars occupy more than half. * At the critical radius $r = 3/2\,r_{S}$ the angular size of the hole reaches 90° (the light■orbit radius $r = 3/2\,r_{S}$ gives $\theta_{\text{crit}} = 90^{\circ}$), meaning the horizon appears to stretch from the vertical to the horizon of the observer’s view. So, visually, a black■hole looks like a dark circular region (the horizon) surrounded by a highly distorted, lensed view of the rest of the universe, with repeated rings of the observer’s own image and background stars wrapped around it.

Sources:

- Page 38, Asimov Black Holes - The Collapsing Universe.pdf
- Page 75, Asimov Black Holes - The Collapsing Universe.pdf
- Page 61, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 85%
